Quantum Computing

Lecture notes from my graduate level Quantum Computing class taught by professor Lijun Chen at the University of Colorado Boulder. This course covers a range of foundational topics in quantum computation and information including: basics to quantum mechanics, entanglement, quantum optimization and machine learning, reliable and scalable quantum computing, and quantum communication and networking. There is no official textbook we are using, but rather a bunch of lecture slides and notes from undergrad and grad courses.

Disclaimer: This document are notes taken by the author (me) a student in graduate school as way to internalize the material I am learning in this class. As such, these notes will inevitably contain mistakes such as errors, typos, etc., personal anecdotes, quips, opinions, and biases. Do what you will with that information, and as a professor once said to me "It's better not to think about it too much".

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1 Basics of Quantum Mechanics

1.1 Bit vs qubit

A bit is the basic unit of computation that takes a value of a 0 or a 1. Think of it as a two-state system: on or off, low or high etc.

The quantum bit (qubit) is a basic unit of information that takes in two-dimensional complex vector space.

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle\,, |a|^2 + |b|^2 = 1$$

- $|0\rangle$ and $|1\rangle$: elements of an orthonormal basis
- $a, b \in \mathbb{C}$ are the probability amplitudes which must satisfy $\sum_i |a|^2 = 1$
- if measured onto the basis, $|0\rangle$ is obtained with a probability $|a|^2$ and $|1\rangle$ with probability $|b|^2$

1.2 Basics of Quantum Mechanics

- how do we represent a quantum state?
- how to represent an observable?
- how does a quantum system evolve over time?
- what operations can we do on a quantum system?
- how do we combine quantum systems?

1.2.1 States

A state is a complete description of a physical system. A state in quantum mechanics is defined as a unit vector in Hibert space, denoted by $|\psi\rangle$ (Dirac's ket notation).

- a vector space over complex numbers $\mathbb{C}$
- **Inner Product**: $\langle\psi|\phi\rangle$ that maps an ordered pair of vectors to $\mathbb{C}$ and that has these properties:
  - Positivity: $\langle\psi|\psi\rangle > 0$ for $|\psi\rangle \neq 0$
  - Skew symmetry: $\langle\psi|\phi\rangle = \langle\phi|\psi\rangle$
- **Normalization**: $\| |\psi\rangle \| = \sqrt{\langle\psi|\psi\rangle}$
- **Global phase**: $|\psi\rangle$ and $e^{ia}\,|\psi\rangle\,, a \in \mathbb{R}$ are indistinguishable

Suppose there are N distinguishable states. A quantum state $|\psi\rangle$ is a *superposition* of these N states:

$$|\psi\rangle = a_1\,|1\rangle + a_2\,|2\rangle \cdots + a_n\,|N\rangle$$

We can think of $|\psi\rangle$ as the unit vector:

$$\begin{bmatrix} a_1 \\ a_2 \\ \cdots \\ a_N \end{bmatrix}$$

With the conjugate transpose:

$$\langle\psi| = (a_0^*, a_1^*, \ldots, a_n^*)$$

So, for the two states $|\psi\rangle = a_n |N\rangle$ and $|\phi\rangle = b_n |B_N\rangle$, their *inner product* is:

$$\langle\psi\rangle \phi = a_1^* b_1 + \cdots + a_n^* b_n$$

The *outer product* projection to $|\psi\rangle$ corresponds to:

$$\begin{bmatrix} a_1 \\ a_2 \\ \cdots \\ a_n \end{bmatrix} [b_1^*, b_2^*, \ldots, b_n^*]$$

1.2.2 Observable

A property of a physical system that can be measured. It's defined mathematically as a self-adjoint operator on Hilbert space

$$A = A^\dagger$$

- can be diagonalized:
$$A = \sum_n a_n E_n$$
with a_n being eigenvalues of A, and E_n is the corresponding orthogonal projection onto the space of eigenvectors with eigenvalues a_n.
- self-adjoint guarantees these eigenvalues a_n are *real* because the measurements represent actual outcomes (not fake ones)
- the projections satisfy: $E_n = E_n^\dagger$ and $E_n E_m = \delta_{n,m} E_n$
What this means is:
  - Hermitian (self-adjoint)
  - $\delta_{n,m}$ is the **Kronecker delta** which equals 1 if $n = m$ and - if $n \neq m$.

  For case $n = m$ (idempotent) applying the same projection twice does nothing.
  For case $n \neq m$ $E_n E_m = 0$. Projecting onto space m and then a different space gives you nothing because the two spaces don't overlap at all. The eigenspaces are mutually perpendicular.

1.2.3 Measurement

The measurement of an observable A prepares an eigenstate $|n\rangle$ of A (state collapse):

- Probability of outcome a_n:

$$Prob(a_n) = \|E\,|\psi\rangle\,\|^2 = \langle\psi|\,E\,|\psi\rangle$$

- Post-measurement state:

$$\frac{E_n\,|\psi\rangle}{\|E_n\,|\psi\rangle\,\|}(\text{state collapse})$$

  Note: when measured again, the outcome remains the same.

- Expected value over many trials: $\sum_n a_n \text{Prob}(a_n) = \langle\psi|\,A\,|\psi\rangle$

Measuring is projecting the state vector onto an eigenspace and normalizing.

1.2.4 Dynamics

Dynamics describe how a state evolves over time. The time evolution of a closed system is described by a *unitary operator*

If the initial state at time t is $|\psi\rangle\,t$, then the final state $|\psi\rangle\,t'$ at time t' can be expressed as

$$|\psi(t')\rangle = U(t',t)\,|\psi(t)\rangle\,|\psi(t)\rangle$$

with unitary operator satisfying

$$U^\dagger U = UU^\dagger = I$$

Since the unitary operator preserves inner products: $|\psi'\rangle = U\,|\psi\rangle$ and $|\phi'\rangle = U\,|\phi\rangle$, then

$$\langle\psi'|\phi'\rangle = \langle\psi|\,U^\dagger U\,|\phi\rangle = \langle\psi|\phi\rangle$$

1.2.5 Composite system

If the Hilbert space of system $\mathcal{A}$ is $\mathcal{H}$ and $\mathcal{B}$ with $\mathcal{H}_\mathcal{B}$, then the Hilbert space of the composite systems AB is the *tensor product* $\mathcal{H}_\mathcal{A} \otimes \mathcal{H}_\mathcal{B}$

$$\mathcal{H}_{\mathcal{AB}} = \mathcal{H}_\mathcal{A} \otimes \mathcal{H}_\mathcal{B}$$

- with dimension $d_\mathcal{A} d_\mathcal{B}$ and orthonormal basis $\{|i\rangle_\mathcal{A} \otimes |j\rangle_\mathcal{B}\}$

1.2.6 Entanglement

A state like the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ cannot be factored as a tensor product $|\psi\rangle_A \otimes |\phi\rangle_B$

1.2.7 Density matrix

When you only care about subsystem A, you compute expectation values
$\langle \psi M_A| \otimes I_B |\psi\rangle = tr(M_A \rho_a)$ which defines the reduced density matrix $\rho_A = tr_B |\psi\rangle \langle\psi|$ the *partial trace*.

Density matrices have the following properties:

- ρ is self-adjoint $\rho = \rho^\dagger$
- ρ is positive i.e. $\langle\psi| \rho |\psi\rangle \geq 0$ for any $|\psi\rangle$
- $tr\rho = 1$
- pure state if $\text{rank}(\rho) = 1$ and mixed state otherwise
- pure state if $\rho^2 = \rho$ and mixed state otherwise

1.2.8 No cloning theorem

States that no arbitrary transformation can make a copy of an arbitrary quantum state.

Take two arbitrary states $|\psi\rangle_A$ and $|\phi\rangle_A$

$$U |\psi\rangle_A |0\rangle_E = |\psi\rangle_A |\psi\rangle_E$$
$$U |\phi\rangle_A |0\rangle_E = |\phi\rangle_A |\phi\rangle_E$$

The unitary transformation preserves the inner product

$$\langle\psi|\phi\rangle = \langle\psi|\phi\rangle^2$$

But $\langle\psi|\phi\rangle = 0$ or 1 which is a contradiction unless the states were already orthogonal or identical. Thus, cloning arbitrary states is algebraically impossible.

2 Quantum Circuits

Quantum circuits are one of the foundational concepts of quantum computing. It just boils down to a procedure that performs unitary operations (quantum gates) on a number of qubits. A circuit is then constructed from a finite number of these gates acting on the qubits. Finally, an orthogonal measurement of all qubits is performed which projects each measured qubit onto the basis of $\{0, 1\}$.

In vector form, the basis in Hilbert space looks like

$$|0\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, |1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Some basic linear operators include the identity, and Pauli X, Y, Z matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \ \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \ \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \ \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

2.1 What is a quantum gate?

A quantum gate is nothing more than a matrix multiplying a vector. It acts similarly like a linear transformation.

Since it is relatively hard to explain with words, examples should work better. A popular example of a quantum gate is the Hadamard gate which maps $|0\rangle$ to $|+\rangle$ and $|1\rangle$ to $|-\rangle$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

Mapping the Hadamard gate to $|0\rangle, |1\rangle$ would look like

$$|+\rangle = H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), |-\rangle = H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

2.1.1 Quantum interference

Quantum interference occurs when the probability amplitudes of quantum states combine to either increase or decrease the likelihood of a specific measurement.

Think of it like a pond and on this pond there are rubber ducks covering the entire pond (that's a lot of rubber ducks). Now, imagine you send a ripple down this pond. The ducks would bob up and down as the ripple passed through them. Quantum interference is like sending a ripple down one end of the pond, then sprinting to a new location and creating another ripple there. Some ripples cancel out, and some get bigger. Meanwhile, all the ducks just sit on the pond, taking the waves of interference as they bob up and down without a care in the world...

Now back to quantum. An example of an interference pattern can be inversely applied using the Hadamard gate

$$
\begin{aligned}
H|+\rangle &= \frac{1}{\sqrt{2}}(H|0\rangle + H|1\rangle) \\
&= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\
&= \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle + \frac{1}{2}|0\rangle - \frac{1}{2}|1\rangle \\
&= |0\rangle \\
H|-\rangle &= \frac{1}{\sqrt{2}}(H|0\rangle - H|1\rangle) \\
&= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) - \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \\
&= \frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle - \frac{1}{2}|0\rangle - \frac{1}{2}|1\rangle \\
&= |1\rangle
\end{aligned}
$$

With the positive Hadamard gate, the $|0\rangle$ both come out to $\frac{1}{2}$ and add up (constructive interference). Meanwhile the $|1\rangle$ both cancel each other out.

2.1.2 Phase shift gate

A phase shift gate $P(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix}$ is a tool used to inject the phase needed to make a cancellation in the right spot.

Example

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Apply the phase change:

$$P(\theta)|+\rangle = (|0\rangle + e^{i\theta}|1\rangle)$$

Apply Hadamard interference

$$HP(\theta)|+\rangle = \frac{1}{2}[(1 + e^{i\theta}|0\rangle) + (1 - e^{i\theta}|1\rangle)]$$

Taking the $|\texttt{amplitude}|^2$ knowing that $|1 + e^{i\theta}|^2 = 2 + 2\cos(\theta)$ of each term gives us

$$
\begin{aligned}
\text{P (measure 0)} &= \frac{1 + \cos\theta}{2} = \cos^2(\frac{\theta}{2}) \\
\text{P (measure 1)} &= \frac{1 - \cos\theta}{2} = \sin^2(\frac{\theta}{2})
\end{aligned}
$$

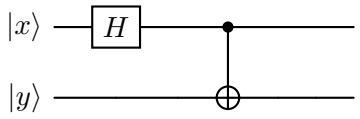
2.2 Multi-qubit gates

2.2.1 The controlled NOT gate (CNOT gate)

Maps $|x, y\rangle$ to $|x, x \oplus y\rangle$.

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Below is an example of applying the Hadamard gate to the first ket and the CNOT gate to the second ket

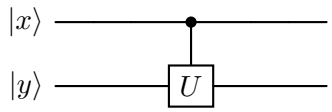


2.2.2 Controlled-U gate

It's defined as $|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U$.

Alright, let's back it up and explain that in plain English: If the control bit is "0", do nothing to the target (apply the identity matrix which does nothing). Else if the target is "1" apply U to the target.a

Here's what it looks like in circuit form



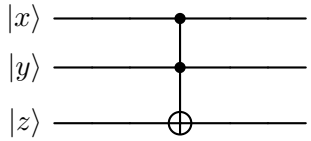
The CNOT gate is this exact operation with the Pauli bit flip.

2.2.3 Toffoli Gate

AKA the CCNOT gate which is the 3-qubit version of the CNOT gate.

It maps $|x, y, z\rangle$ to $|x, y, z, \oplus(x \wedge y)\rangle$.

The CCNOT gate is a universal gate for reversible classical computing. Any classical reversible circuit can be constructed from it.



2.3 The quantum circuit model

$$|\psi\rangle = \sum_{x \in \{0,1\}^n} a_x |x\rangle \to U |\psi\rangle$$

Instead of building a really big $2^n \times 2^n$ matrix, we can decompose it into a series of smaller operations acting on a few qubits at a time to not overwhelm ourselves!

Building a circuit has four parts:

- **Initialization**: $|x\rangle = |0\rangle^{\otimes n} = |000\ldots0\rangle$
  - Start in the lowest energy state of quantum register
- **A universal set of quantum gates** $\{U_1, U_2, \ldots U_m\}$
  - A finite set of gates can be used to approximate n-qubits to a desired accuracy
- **Classical control**: A classical computer builds a circuit and then executes it in order. (Later: see the quantum-classical paradigm)
- **Measurement** in the standard basis $\{0, 1\}$.
  - Complexity of complicated measurement accounted in complexity of an algorithm.
  - Delayed measurement vs measurement at intermediate stages.
    - delayed measurement is general
    - real devices often measure in between based off of outcome and branch

2.4 Universal Gates

A gate set is universal if some circuit V built from it can approximate any unitary transformation U on any number of qubits and get a desired accuracy ϵ.

$$\|V - e^{i\theta}U\|_\infty \leq \epsilon$$

There are two flavors:

1. **exact universality**: requires uncountable gate set; e.g., the set of two qubit gates, any entangling two-qubit gate plus the set of one-qubit gate
2. **generic universality**: almost any two-qubit gate is universal if it can be applied to any pair of qubits.
3. examples: $\{\text{CCNOT}, H, P(\pi/2)\}$, $\{H, \text{control} - e^{i\pi/4}P(\pi/2)\}$

2.5 Solovay-Kitaev theorem

This theorem answers these two questions:

- how large does quantum circuit V need to be to approximate a given unitary transformation U to a desired accuracy?
- how large of a classical circuit is needed to find such a quantum circuit?

Turns out the answer to both questions is: given a universal set of quantum gates, the quantum circuit and classical circuit need to be $\log^{O(1)1/\epsilon}$ large.

3 Quantum Algorithms

3.1 Deutsch's Algorithm

Suppose you have a black-box function f mapping a single bit to another bit, and you want to determine if f is a constant function or a balanced function. In classical computation, this requires two computations. But with quantum mechanics, we can do it with one!

There are four possible functions:

	f_0	f_1	f_2	f_3
0	0	0	1	1
1	1	0	1	0

Q: Why can't we do it with a single qubit?

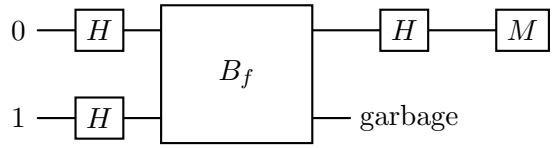
A single qubit acting on $|0\rangle$ *and* $|1\rangle$ would be $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ which is not a unitary matrix.

With a two qubit gate, we can use unitary B_f (a "phase oracle" of f) acting as:

$$B_f : |x\rangle\,|y\rangle \to |x\rangle\,|y \oplus f(x)\rangle$$

...which is always unitary because it's XOR-ing it's own inverse.

Example



Let's walk through the above circuit.

Before applying the Hadamard gate we have

$$|0\rangle\,|1\rangle$$

Applying the Hadamard to both yields

$$|0\rangle\,|1\rangle = \frac{1}{2}\Big[\,|0\rangle\,(|0\rangle - |1\rangle) + |1\rangle\,(|0\rangle - |1\rangle)\Big]$$

Expanding and applying B_f (which turns $|x\rangle\,|y\rangle$ into $|x\rangle\,|y \oplus f(x)\rangle$, and flipping y inside $(|0\rangle - |1\rangle)$ just flips its overall sign when $f(x) = 1$) gives:

$$\frac{1}{2}\Big[\,|0\rangle\,|f(0) \oplus 0\rangle - |0\rangle\,|f(0) \oplus |1\rangle\rangle + |1\rangle\,|f(1) \oplus 0\rangle - |1\rangle\,|f(1) \oplus |1\rangle\rangle\,\Big]$$

This can be simplified as

$$= \frac{1}{2}(-1)^{f(0)}|0\rangle(|0\rangle - |1\rangle) + \frac{1}{2}(-1)^{f(1)}|1\rangle(|0\rangle - |1\rangle)$$
$$= \Big(\frac{1}{\sqrt{2}}(-1)^{f(0)}|0\rangle + \frac{1}{\sqrt{2}}(-1)^{f(1)}|1\rangle\Big)\Big(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle\Big)$$

Notice how the state of the second qubit has not changed! This is known as "phase kickback".

We can now throw out the second qubit which leaves us with

$$\Big(\frac{1}{\sqrt{2}}(-1)^{f(0)}|0\rangle + \frac{1}{\sqrt{2}}(-1)^{f(1)}|1\rangle\Big)$$

which can be written as

$$(-1)^{f(0)}\Big(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}(-1)^{f(0)\oplus f(1)}|1\rangle\Big)$$

The final Hadamard gate takes us to

$$(-1)^{f(0)}|f(0)\oplus f(1)\rangle$$

3.2 Grovers algorithm

Suppose you have a key hidden among various boxes and you want to find the box with that specific key. Classically, in the worst case, finding this would require you to iterate over every single piece of hay. With quantum mechanics, we can lower that complexity and find it in $O(\sqrt{2^n})$ times.

A more formal way of explaining this is given a function $f : \{0,1\}^n \to \{0,1\}$ with

$$f(x) = \begin{cases} 0 \; x \neq w \\ 1x = w \end{cases}$$

for some unknown w, find w.

3.2.1 Setup

The function $f : \{0,1\}^n \to \{0,1\}$ implemented with the unitary transformation

$$U_f|x\rangle|a\rangle = |x\rangle|a \oplus f(x)\rangle$$

where $x \in \{0,1\}^n$ and $a \in \{0,1\}$

The oracle U_f turns into a *phase oracle* which is a quantum black-box operation that marks a specific target state by multiplying them by a negative sign or a phase factor $(-1)^{f(x)}$.

$$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$$

where $(-1)^{f(x)}$ equals 1 if $x \neq w$ and -1 if $x = w$. This is essentially saying that we leave every box alone except for the one we want the sign to flip.

Ignoring the output register we can express U_w as

$$U_w = I - 2\left|w\right\rangle\left\langle w\right|$$

Given n-qubit state we can express this as

$$U_f = \left|\psi\right\rangle = a\left|w\right\rangle + b\left|w^{\perp}\right\rangle \rightarrow -a\left|w\right\rangle + b\left|w^{\perp}\right\rangle$$

The unitary operator induces a reflection of the vector $|\psi\rangle$ about the hyperplane created by $\langle w|w^{\perp}\rangle$.

3.2.2 Walkthrough

First, we prepare a uniform superposition $|s\rangle = H^{\otimes n}\left|000\ldots0\right\rangle = \frac{1}{\sqrt{2^n}}\sum_{x\in\{0,1\}^n}\left|x\right\rangle$.

Next, we apply the *Grover iteration* in succession where each iteration enhances the value of the secret key while suppressing all other values.

$$U_G = U_s U_f$$

U_f is the query and U_s is the reflection about the "axis" $|s\rangle$ which can be expressed as

$$U_s = 2\left|s\right\rangle\left\langle s\right| - I$$

So what does U_G do?

It preserves the plane spanned by $|s\rangle$ and $|w\rangle$, so we can focus solely on the plane. First U_f reflects $|s\rangle$ about the axis $|w^{\perp}\rangle$.

Then U_s reflects $U_w\left|s\right\rangle$ about the axis $|s\rangle$.

The effect of U_G is a counterclockwise rotation of the state vector by 2θ where θ is the initial angle between $|s\rangle$ and $|w\rangle$.

The rotation of the initial angle θ is given by

$$\sin(\theta) = \langle s|w\rangle = \frac{1}{\sqrt{2^N}}$$

The required number of Grover iterations $T = O(\sqrt{2^n})$. If we repeat it T times, the vector is rotated away from the $|w^{\perp}\rangle$ $(2T+1)\theta$.

We may choose T s.t. $(2T+1)\theta = \frac{\pi}{2} + \delta$ with $|\delta| \leq 0$.

The probability to find $|w\rangle = P(w) = \cos^2\delta \geq 1 - \delta^2 \geq 1 - \frac{1}{2^n}$.

Suppose there are r marked states where r is known. We can define the marked states as

$$|\text{marked}\rangle = \frac{1}{\sqrt{r}} \sum_{i=1}^{r} |w_i\rangle \,, \sin\theta = \langle s|\text{marked}\rangle = \sqrt{r/2^n}$$

and the Grover iteration rotates by 2θ in the plane spanned by $|s\rangle$ and $|\text{marked}\rangle$.

Classically, this problem would need $O(2^n/r)$ queries. Quantumly, we would need $T = O(\sqrt{\frac{2^n}{r}})$ queries (our quadratic speedup).

What if r is not known a priori? The success probability oscillates with the period $\sim \pi/2\sqrt{2^n/r}$. If we choose T uniformly at random between 0 and $\sim \pi/4\sqrt{2^n}$, then if there is a solution ($r > 0$), a solution will be found with a probability greater than $1/2$ if it exists.

3.3 Quantum Fourier Transformation

The classical Fourier Transform implementation

$$\sum_x f(x)\,|x\rangle \to \sum_y \Big(\frac{1}{\sqrt{M}} \sum_x e^{i2\pi xy/M} f(x)\Big)\,|y\rangle$$

is multiplication by an $M \times M$ matrix where the (x, y) matrix element is $\left(e^{2\pi i/M}\right)^{xy}$ which naively requires $O(M^2)$ elementary operations. We can reduce the number of operations to $O(M \log M)$, by assuming that $M = 2^m$, we can express x and y as binary expansions.

$$x = x_{m-1} \cdot 2^{m-1} + x_{m-2} \cdot 2^{m-2} \cdots + x_1 \cdot 2 + x_0$$
$$y = y_{m-1} \cdot 2^{m-1} + y_{m-2} \cdot 2^{m-2} \cdots + y_1 \cdot 2 + y_0$$

We can throw away terms containing m or more powers of 2, hence

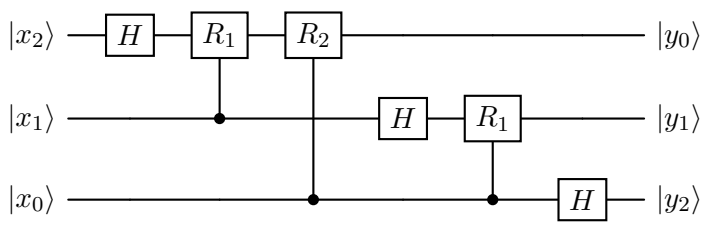
$$\frac{xy}{2^m} \equiv y_{m-1}(.x_0) + y_{m-2}(.x_1x_0) + \cdots + y_0(.x_{m-1}x_{m-2}\cdots x_0)(mod\ 1)$$

where the factors in parentheses are binary expansions e.g.

$$.x_2x_1x_0 = x_2/2 + x_1/2^2 + x_0/2^3$$

QFT takes each computational basis to an *unentangled* state of n qubits.

Let's consider the case $n = 3$. The circuit would look like



note the order of the bits are reversed in the output. Each Hadamard gate acts as

$$H : |x_k\rangle \rightarrow \frac{1}{\sqrt{2}}\Big(|0\rangle + e^{2i(.x_k)} |1\rangle \Big)$$

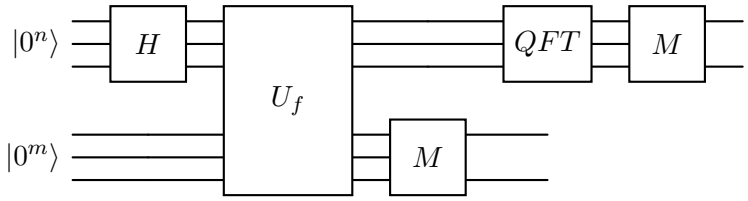
The two-qubit conditional rotations are defined as

$$R_d = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2^d} \end{pmatrix}$$

where d is the "distance" between the qubits.

For the $n = 3$ qubit state we can see that the QFT is constructed of three Hadamard gates and three controlled-R gates. For a general n, we require n Hadamard gates and $n(n-1)/2$ controlled R gates. A two qubit gate is applied to each pair of qubits with controlled relative phase $\pi/2^d$ where d is the distance between the qubits. Thus, the circuit that implements QFT has the size $(\log M)^2$.

3.4 Period finding



Imagine a quantum oracle that computes a function

$$f : \{0,1\}^n \rightarrow \{0,1\}^m$$

that has an unknown period r where r is a positive integer satisfying

$$1 \ll r \ll 2^m \ll 2^n$$

The quantum oracle is expressed as

$$f(x) = f(x + kr)$$

where $k \in \mathbb{Z}_+$ such that x and $x + kr$ lie in $\{0, 1, 2, \ldots, 2^n - 1\}$. Classically, this is a hard problem. If $r = O(2^{n/2})$, then we would need $O(2^{n/4})$ queries before finding two x values with the same function value. Quantumly, period r can be found in $poly(n)$ time using the quantum Fourier Transformation.

3.4.1 Finding the period

Given a function $f : \{0,1\}^n \to \{0,1\}^m$ implemented by the unitary transformation U_f

$$U_f |x\rangle |a\rangle = |x\rangle |a \oplus f(x)\rangle$$

where $x \in \{0,1\}^n, a \in \{0,1\}^m$ and bitwise XOR.

We prepare the uniform superposition over x, query f, and measure the *output* register. This collapses the input register onto a superposition of just the x's consistent with the observed ouput. And because f is periodic, it is an *evenly spaced comb* of L values.

How it works

Before measuring we have two registers called $|0^n\rangle$ and $|0^m\rangle$.

We add the Hadamard gates and apply it to the first register.

$$|0^n\rangle |0^m\rangle \Rightarrow \Big(\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle\Big) |0^m\rangle \Rightarrow \Big(\frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle\Big) |f(x)\rangle$$

Then we can measure the output register obtaining the result $|f(x_0)\rangle$ for some $0 \le x_0 \le r$. This measurement prepares the input register for the state

$$\frac{1}{\sqrt{L}} \sum_{j=0}^{L-1} |x_0 + jr\rangle$$

where L is the number of x values with the function value $f(x_0)$ and $2^n - r \le x_0 + (L-1)r < 2^n$.

We need to extract the value of r from the prepared state. We apply the QFT and then measure by projecting onto the computation basis.

Applying QFT to obtain

$$\frac{1}{\sqrt{2^n L}} \sum_{y=0}^{2^n-1} e^{i2\pi x_0 y/2^n} \sum_{j=0}^{L-1} e^{i2\pi jry/2^n} |y\rangle$$

If we now measure the computation basis, the probability is

$$Prob(y) = \frac{L}{2^n} \Big| \frac{1}{L} \sum_{j=0}^{L-1} e^{i2\pi jry/2^n} \Big|^2$$

This is independent of x_0, and strongly favors values of y s.t. $yr/2^n$ is close to an integer.

Special case

If $2^n/r$ happens to be an integer, then $2^n/r = L$ and

$$Prob(y) = \frac{1}{r}\Big|\frac{1}{L}\sum_{j=1}^{L-1} e^{i2\pi jy/L}\Big|^2 = \begin{cases} \frac{1}{r} & y = \text{ a multiple of } L \\ 0 & \text{otherwise} \end{cases}$$

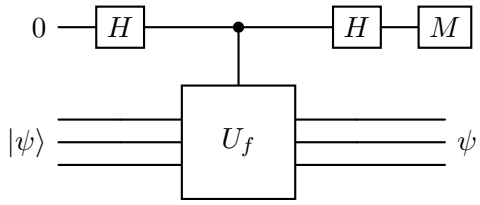
Constructive interference if y is a multiple of $2^n/r$ and destructive otherwise.

3.5 Phase estimation

Let U be a unitary transformation that has an orthonormal basis of eigenvectors $|\psi_1\rangle, |\psi_2\rangle \ldots |\psi_N\rangle$ with associated values $e_1^{2\pi i\theta}, e_2^{2\pi i\theta}, \ldots, e_N^{2\pi i\theta}$.

Given a unitary transformation U and one of its eigenvector $|\psi_j\rangle$, find the corresponding eigenvalue, θ_j. That is the phase estimation problem.

How it works



First apply the Hadamard gate

$$|0\rangle \Rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \Rightarrow \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\theta}|1\rangle) \Rightarrow \frac{1}{2}(1 + e^{2\pi i\theta}|0\rangle) + \frac{1}{2}(1 - e^{2\pi i\theta}|1\rangle)$$

After the unitary transformation, the eigenvector remains unchanged while we have been able to put $e^{2\pi i\theta}$ into the phase of the first qubit will transform this information into the amplitude which we will be able to measure.

Let $P(0)$ and $P(1)$ be the probability of seeing a 0 or 1 respectively on measuring the first qubit (kind of like Deutsch's). The probabilities will be

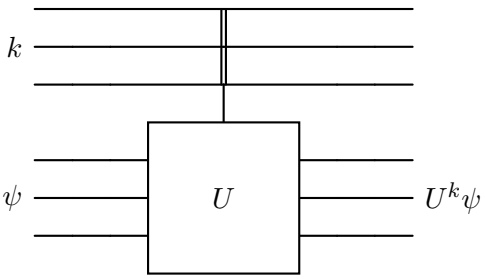
$$\text{Prob}(0) = |\frac{1}{2}(1 + e^{2\pi i\theta})|^2 = \frac{1 + \cos(2\pi\theta)}{2}$$
$$\text{Prob}(1) = |\frac{1}{2}(1 - e^{2\pi i\theta})|^2 = \frac{1 - \cos(2\pi\theta)}{2}$$

This estimates $\cos(2\pi\theta)$ to m bits of precision, but only after repeating $O(2^m)$ times — not really efficient by itself.

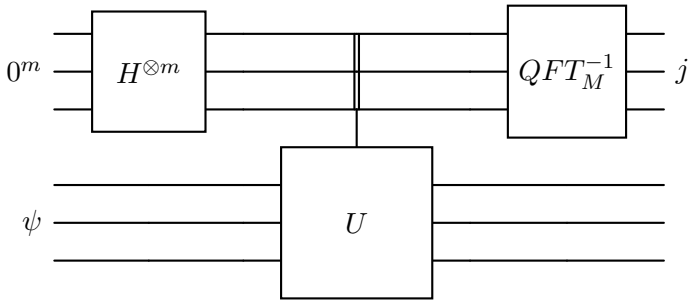
Efficient way

Let's make this more efficient. Suppose we can implement the unitary transformation $C_m - U$ transformation defined below.

For any unitary transformation U, let $C_k - U$ stand for a "k-controlled U" circuit which implements the transformation $|k\rangle \otimes |\psi\rangle \rightarrow |k\rangle \otimes U^k|\psi\rangle$ as shown below



Estimating θ within m-bit accuracy is equivalent to estimate integer j where $\frac{j}{2^m}$ is the closest approximation to θ. Let $M = 2^m$ and $w_m = e^{\frac{2\pi i}{M}}$



The circuit above follows the sequence of transformations

$$
\begin{aligned}
|0^m\rangle |\psi\rangle &\xrightarrow{H^{\otimes M}} \Big(\frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} |k\rangle \Big) \otimes |\psi\rangle \\
&\xrightarrow{C_m - U} \Big(\frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} e^{2\pi i \theta k} |k\rangle \Big) \otimes |\psi\rangle \\
&= \Big(\frac{1}{\sqrt{M}} \sum_{k=0}^{M-1} e^{2\pi i k(\theta - \frac{j}{M})} |k\rangle \Big) \otimes |\psi\rangle
\end{aligned}
$$

Note that the first register now contains the Fourier Transform mod M of j and if we apply the reverse Fourier Transformation mod M we get back j.

The probabilities are as follows:

- $Prob(j) = 1$ if $\theta = \frac{j}{M}$
- $Prob(j) \geq 4/\pi^2$ if $\frac{j}{M}$ is the closest approximation

3.6 Factoring

Suppose we have a positive integer N, find a prime factorization such that $N = p_1^{k_1} p_2^{k_2}, \cdots, p_m^{k_m}$ where p_i's are prime numbers and k_i's are positive integers.

This is believed to be a hard problem, classically. and its hardness is the basis of RSA encryption scheme for public key cryptography.

Order finding

For a random $a < N$ coprime to N (having no common factor with N), its multiplicative *order* r is the smallest positive integer with

$$a^r \equiv 1 \pmod{N}$$

The order of a mod N is the period of the function

$$f_{N,a} = a^x \pmod{N}$$

We know that there's an efficient quantum algorithm that can find the period of a function. Therefore, we can compute $f_{N,a}$ efficiently if we can find the order of a efficiently.

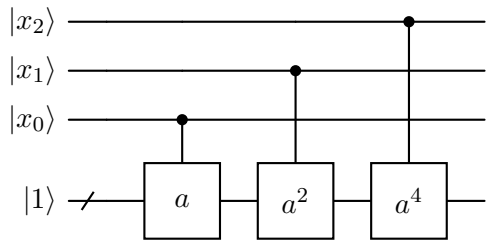
If $x < 2^m$ and we express x as a binary expansion

$$x = x_{m-1} \cdot 2^{m-1} + x_{m-2} \cdot 2^{m-2} + \cdots + x_0$$

we have

$$a^x \pmod{N} = (a^{2^{m-1}})^{x_{m-1}} (a^{2^{m-2}})^{x_{m-2}} \ldots (a)_0^x \bmod N$$

Therefore, finding the order of a is a period-finding problem, and a^x mod N can be computed in $O(\log N)^3$ via the repeated squaring: compute $a, a^2, a^4, a^8, \ldots a^{2^k}$.



3.6.1 Shor's algorithm

- pick a random $a < N$
- if the $GCD(a, N) \neq 1$, then $GCD(a, N)$ is nontrivial factor of N (done!)
- if $GCD(a, N) = 1$, find the order r of a via quantum period finding
- if r is even, N divides $(a^{r/2} + 1)(a^{r/2} - 1)$ and
  - either N divides $a^{r/2} + 1$
  - or, factors of N split between $a^{r/2} + 1$ and $a^{r/2} - 1$ and then $GCD(a^{r/2} \pm 1, N)$ are nontrivial factors of N (done!)

3.6.2 Factoring via phase estimation

Instead of framing it like period finding, define $U_a |x\rangle = |ax \bmod N\rangle$. Since a has order r, $U_r^a = I$, so its eigenvalues are exactly the r-th roots of unity $e^{i2\pi k/r}$ for $k = 0, \ldots, r-1$. The phase estimation determines the value of k/r, which is the same as period finding.

4 Quantum Complexity Theory

4.1 What is a solvable problem?

Which problems are solvable in practice? How do we rate how hard a problem is to solve? We define solvable problems as those that are able to be solved in polynomial time. Classically this includes: shortest path, min cut, 2-satisfiability, primality testing, etc. Some classical problems that are believed to be not solvable classically are: longest path, max cut, 3-satisfiability, factoring, to name a few.

4.2 Complexity theory

We can classify hardness of a problem in terms of the resources needed to solve large instances of these problems. The computational efficiency can be measured by scaling the resources (time/space) of the algorithm.

4.3 Polynomial time reduction

problem X polynomial-time-reduces to problem Y if arbitrary instances of problem X can be solved using:

- polynomial number of basic computational steps
- polynomial number of calls to oracle that solves problem Y

We note this as $X \leq_p Y$.

We can use polynomial time reduction in these ways

- design algorithms: if $X \leq_p Y$, Y's polynomial time solvable implies X is polynomial time-solvable.
- establish intractability: if $X \leq_p Y$ and X is not polynomial time solvable, Y cannot be polynomial time solvable.
- establish equivalence: X and Y are equivalent if both $X \leq_p Y$ and $Y \leq_p X$ denoted $X \equiv_p Y$.

4.4 Different types of problems

We can classify different types of problems into three types:

- decision problems: given a question, an input and a threshold w, determine the existence of a solution at least as good as w, answering YES if such a solution exists and NO otherwise.
- search problems: given a question, an input and a threshold w, find a solution at least as good as w.

- optimization problems: given a question and an input, find the best solution.